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Roll No

EC-702 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Satellite Communication
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. Define the following parameters with reference to satellite orbits :
 - i) Apogee and perigee
 - ii) Eccentricity
 - iii) Ascending and descending nodes
2. Discuss briefly about the following
 - a) Station keeping
 - b) Thermal control
 - c) TT and C sub system
3. Explain altitude and orbit control technique used in satellite.
4. Explain how intermodulation noise originates in a satellite link and how it can be reduced?
5. What is meant by C/N ratio? The C/N value for a satellite circuit are uplink 25 dB and downlink 15 dB. Calculate the overall C/N value.
6. Obtain the expression for the distance of the satellite from the earth and evaluate the distance for geo-synchronous satellite.

OR

Determine combine uplink downlink C/N ratio.
7. Give introductory words on Satellite systems. Discuss the frequency allocations in satellite systems.

OR

Discuss about BDS TV system design and link budget.
8. Write short notes (any four)
 - a) VSAT
 - b) DBS TV
 - c) Transponder
 - d) Eccentricity
 - e) Anik satellites
 - f) Morelos and Satmex – 5
